MLAX Player Wellness Survey Analysis - 2025 (Full Season)

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Executive Summary

This report presents a comprehensive analysis of daily wellness check-in data collected from the men's lacrosse team during the 2025 season. The primary objective is to assess player wellness using five known performance indicators (KPIs): **sleep quality, motivation, overall stress, physical fatigue, and soreness**.

The analysis distinguishes primary drivers, secondary drivers, and supporting indicators of wellness, while identifying meaningful differences across playing positions.

Key Findings:

1. Primary Drivers - **Strongest, most consistent statistical relationships with Wellness scores**
    - **Sleep Quality**: Players reporting better sleep quality consistently show higher overall wellness scores, making it a critical factor for performance and recovery.
    - **Motivation**: Higher motivation levels are strongly associated with better wellness scores, suggesting that psychological factors play a key role in athletes' overall health.
    - **Overall Stress**: Elevated stress levels are linked to lower wellness scores, particularly among certain positions, indicating that stress management may be an important area for intervention.
2. Secondary Drivers - **Statistically significant but slightly less dominant than primary drivers**
    - **Physical Fatigue**: Higher physical fatigue is associated with lower wellness scores, especially for positions with greater physical demands, highlighting the need for tailored recovery strategies.
    - **Soreness**: While soreness is related to wellness, it appears to be a less consistent predictor than sleep and motivation, suggesting it may be more of a symptom than a primary driver.
3. Supporting Indicators - **Contextual or moderating factors rather than standalone drivers**
    - **Position-Specific Patterns**: Certain positions (e.g., FO) show systematically lower wellness scores, which may reflect unique role demands or recovery challenges that require targeted support.
    - **Day of Week Effects**: Wellness scores vary by day of the week, with midweek check-ins reflecting training stress and recovery dynamics, while pre-game and post-game status may require separate assessment.

Position-Specific Findings:

1. Motivation has a much stronger relationship with wellness for Defense than for any other position. FO and OM players show greater representation in the lower motivation categories.

2. GK and D players are more likely to report Good or Excellent sleep quality. FO players show a higher concentration in the lower sleep quality categories.

3. Most players across all positions report soreness levels in the Moderate or Minimal categories.
4. FO and OM players show relatively higher representation in the higher fatigue categories. DM players are more frequently represented in the lower fatigue categories.
5. FO and A players show a higher concentration in the higher stress categories. DM, OM, and GK players are more concentrated in the lower stress categories.

Practical Implications:

1. **Targeted Interventions**

Focus on improving wellness factors through individualized support, and recovery protocols tailored to specific demands.

2. **Response Rate Improvement**

Implement strategies to increase survey participation, especially on days with lower response rates, to ensure more comprehensive data collection and accurate monitoring of player wellness. This will also give us more data to analyze trends across the week and identify critical periods for intervention.

3. **Single day scores should not be over-interpreted**

Wellness scores can fluctuate day-to-day based on training loads, recovery, and other factors. While certain patterns emerge at the group level, individual scores should be interpreted in context rather than as definitive indicators of performance or health on any given day. This underscores the importance of consistent monitoring and looking at trends over time rather than isolated data points.

Recommendations:

1. **- Increase the frequency of wellness surveys to capture more data across the week to address data gap.**
   - Tuesday's wellness scores will be our baseline scores for the week.
   - Thursday will inform adjustments to training loads and recovery strategies later in the week.
   - Friday surveys could assess pre-game readiness.
   - Sunday surveys could assess post-game recovery.
2. **- Use the change of individual wellness scores from Tuesday to Thursday as the primary inflection point.**

- This change in the individual wellness score will flag athletes needing targeted interventions

3. **- Implement Athlete-Specific Z-Score Monitoring.**

- Athlete readiness will be monitored using individualized Z-scores to flag meaningful drops from normal levels and support timely adjustments to training and recovery plans.

4. **- Educate athletes on the importance of wellness monitoring and self-care practices.**

- Provide resources and support to help athletes understand how sleep, motivation, stress, fatigue, and soreness impact their performance and recovery, empowering them to take proactive steps to optimize their wellness.

5. **- Prioritize primary drivers**

- This includes sleep quality, motivation, stress in wellness interventions, while also addressing secondary drivers fatigue, soreness, and position-specific needs.

6. **- Tracking additional factors that may effect wellness**

- Additional question added to the survey to track illness. This will help us understand how illness impacts wellness scores and recovery dynamics, and allow us to adjust training and recovery strategies accordingly.

Methodology

1. Data Preparation

This analysis uses daily wellness check-in data to produce accurate, consistent, and actionable insights for the team. The data was standardized for quality control, and all player information was anonymized to ensure confidentiality.

2. Data Transformation and Analysis

Wellness responses (stress, fatigue, soreness, sleep quality, and motivation) were converted into standardized numerical scales to allow consistent comparison across athletes and positions. Overall wellness scores were categorized into low, medium, and high levels based on relative team distributions rather than arbitrary cutoffs. Results were weighted to reflect the true positional makeup of the roster, ensuring fair representation. Bootstrapping methods were used to account for variability and strengthen the reliability of the findings.

Together, this approach transforms daily check-ins into a balanced, data-driven framework for monitoring athlete wellness and informing performance decisions.

3. Survey Design

This analysis adjusts for differences in survey participation across positions to ensure fair representation. Results are weighted to reflect the team's actual roster composition, so each position contributes proportionally to the findings.

This ensures that conclusions about player wellness accurately represents the full team, not just the athletes who responded most frequently.

Key Findings

4. Position-Level Wellness Patterns

This step focuses on understanding how reliable the wellness estimates are, not just their averages by using a technique called bootstrapping.

See Appendix A for the detailed results for each wellness indicator by position.

Summary

- **Overall Stress**: Measures perceived stress levels
  - There are positional differences in perceived stress.
  - These estimates reflect average differences across positions and do not imply individual-level stress experiences
- **Physical Fatigue**: Measures perceived physical tiredness and exertion
  - DM players report the highest levels of physical fatigue (95% CI: 3.69–3.89).
  - FO players report the lowest levels of physical fatigue (95% CI: 3.01–3.36).
  - Limited overlap between the DM and FO confidence intervals suggests a meaningful positional difference in fatigue.

- **Soreness**: Measures perceived muscle soreness or bodily discomfort
  - Soreness levels are relatively similar across positions, with substantial overlap in confidence intervals.
  - While DM and FO players show slightly higher soreness estimates, differences across positions appear modest.
  - This suggests that soreness may not differ meaningfully by position alone.
- **Sleep Quality**: Measures perceived restfulness and sleep effectiveness
  - GK players report the highest sleep quality (95% CI: 3.87–4.33).
  - FO players report the lowest sleep quality (95% CI: 2.90–3.30).
  - The non-overlapping confidence intervals between GK and FO indicate a statistically meaningful difference in sleep quality.
- **Overall Wellbeing Score**: Composite measure of physical, mental, and recovery-related wellness
  - DM, GK, and D players report the highest overall wellness scores.
  - FO players report the lowest overall wellness scores (95% CI: 31.46–33.45).
  - The confidence interval for FO is notably lower than those for DM and GK, suggesting meaningful differences in overall wellness scores across positions.

5. Chi-Square Test - Wellness Indicators by Mlax_Position

This section of the analysis examines whether wellness patterns differ systematically by lacrosse position.

See Appendix B for the detailed results of the chi-square tests for each wellness indicator by position.

- **Overall Stress**: Assesses the distribution of perceived stress levels across positions
  - The distribution of overall stress differs across lacrosse positions, as shown by the survey-weighted chi-square test.
  - FO and A players show a higher concentration in the higher stress categories relative to other positions.
  - DM, OM, and GK players are more concentrated in the lower stress categories (Low and Very Low).
  - These results indicate an association between position and perceived stress, but do not imply that position causes stress differences.
- **Physical Fatigue**: Assesses perceived levels of physical tiredness
  - The distribution of physical fatigue varies across positions according to the survey-weighted chi-square test.
  - DM players are more frequently represented in the lower fatigue categories (Low and Very Low).

  - FO and OM players show relatively higher representation in the higher fatigue categories (High and Very High).
  - Observed differences reflect population-level associations rather than individual-level fatigue experiences.
- **Soreness**: Assesses perceived bodily soreness after collapsing extreme categories
  - Soreness levels are broadly similar across positions, with substantial overlap in category distributions.
  - Most players across all positions report soreness levels in the Moderate or Minimal categories.
- **Sleep Quality**: Assesses perceived sleep effectiveness and restfulness
  - Sleep quality distributions differ across positions based on the survey-weighted chi-square test.
  - GK and D players are more likely to report Good or Excellent sleep quality.
  - FO players show a higher concentration in the lower sleep quality categories, including Restless sleep.
  - These findings indicate an association between position and sleep quality, without implying causation.
- **Motivation**: Assesses motivation levels after collapsing low-frequency categories
  - Motivation levels vary across positions, as indicated by the survey-weighted chi-square test.
  - DM, GK, and D players are most concentrated in the Very High motivation category.
  - FO and OM players show greater representation in the lower motivation categories relative to other positions.
- **Overall Wellbeing**: Assesses composite wellness status
  - The distribution of overall wellness differs meaningfully across positions.
  - DM and GK players are more likely to fall in the High wellness category.
  - FO players are disproportionately represented in the Low wellness category.
- **Design-Based Interpretation Cautions**
  - Unmeasured factors such as playing time, workload, recovery access, and role expectations may contribute to observed patterns.
  - Results describe group-level tendencies and should not be generalized to individual athletes.

See Appendix C for the detailed results of the chi-square test p-value for wellness indicators by MLAX position

Survey-weighted chi-square tests indicate that wellness patterns vary meaningfully across playing positions. Statistically significant associations were observed between position and each wellness indicator, including overall stress, physical fatigue, soreness, sleep quality, motivation, and overall wellness. These findings suggest that athletes in different roles may experience distinct physical and psychological demands throughout the season. For example, certain positions appear more concentrated in higher stress or fatigue categories, while others report stronger sleep quality or motivation.

Importantly, these results reflect group-level patterns rather than causal relationships, and unmeasured factors such as workload, playing time, or recovery access may also contribute to the observed differences. Overall, the findings indicate that wellness responses are not uniform across the roster and suggest potential value in monitoring and recovery strategies that account for positional demands.

6. Chi-Square Test - Wellness Indicators by Wellbeing Score

See Appendix D for the detailed results of the chi-square test p-values for wellness indicators by Wellbeing Score

Survey-weighted analyses show that overall wellness is strongly associated with each of the five wellness indicators measured in the daily check-ins: stress, physical fatigue, soreness, sleep quality, and motivation. Athletes reporting higher wellness scores tend to experience lower levels of stress, fatigue, and soreness, while also reporting better sleep quality and higher motivation. These results indicate that player wellness reflects the combined influence of multiple physical and psychological factors rather than a single isolated measure. In practical terms, this supports the use of a holistic wellness monitoring approach, where sleep, recovery, physical load, and psychological readiness are considered together when evaluating athlete health and performance.

7. Weekly Wellness Trends: Visualization and Interpretation

Table 1: Number of Responses by Day of the Week

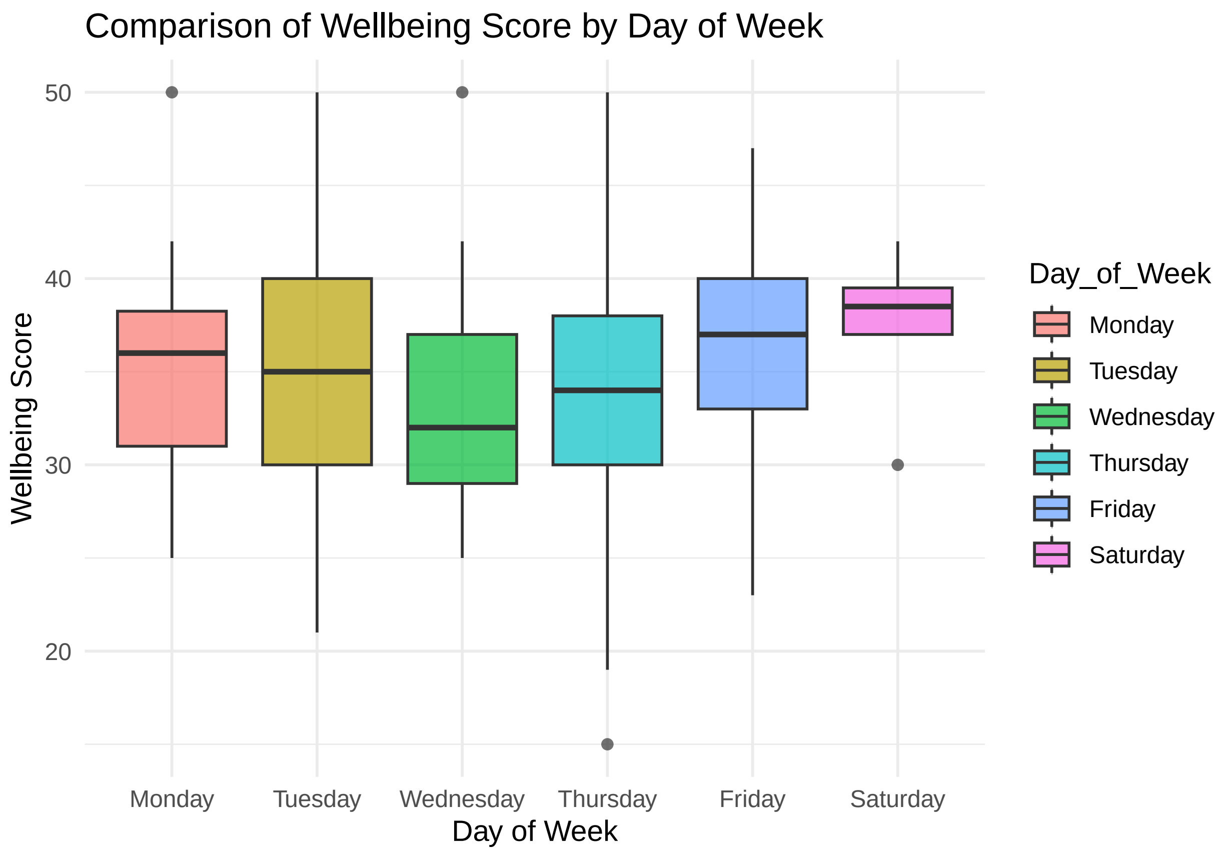
Day_of_Week	Count_of_Day
Sunday	0
Monday	36
Tuesday	501
Wednesday	38
Thursday	450
Friday	21
Saturday	8
Total	**1054**

Table 2: Percentage of Responses by Day of the Week

Day_of_Week	Percentage_of_Responses
Sunday	0.00
Monday	3.42
Tuesday	47.53
Wednesday	3.61
Thursday	42.69
Friday	1.99
Saturday	0.76

Summary Importantly, the distribution of survey responses is heavily concentrated on Tuesday (47.5%) and Thursday (42.7%), meaning the results primarily reflect midweek training stress and recovery dynamics, rather than pre-game or post-game states.

Visualizations of Wellbeing Score by Day of the Week The plot captures how athletes carry fatigue and recovery across the competitive week, especially in relation to Saturday games and subsequent training loads. This box plot emphasizes the player to player spread in wellness scores. It takes a micro look at our wellness scores. This is useful in identifying individual deviations.



Day by Day Interpretation

- **Tuesday** - 47.5% of responses
- Tuesday accounts for nearly half of all responses
- Scores show high variability, indicating uneven recovery.
- This suggests Tuesday may represent a physiological tipping point, where some athletes are adapting well while others accumulate stress.
- These results reflect carryover fatigue for certain athletes.
- **Thursday** - 42.7% of responses
- Wellness scores variability improve relative to Tuesday, indicating partial recovery or training adjustments.
- Variability narrows slightly, suggesting more consistent readiness across the roster
- However, some athletes still show lower scores, indicating incomplete recovery for a subset of players.
- Differences between players become more visible, highlighting individual recovery needs.

- **All other days** <10% of response
- Limits interpretability
- Sunday and Monday data are too sparse to draw conclusions about immediate post-game recovery.
- Friday and Saturday data are also limited, restricting insights into game day dynamics.

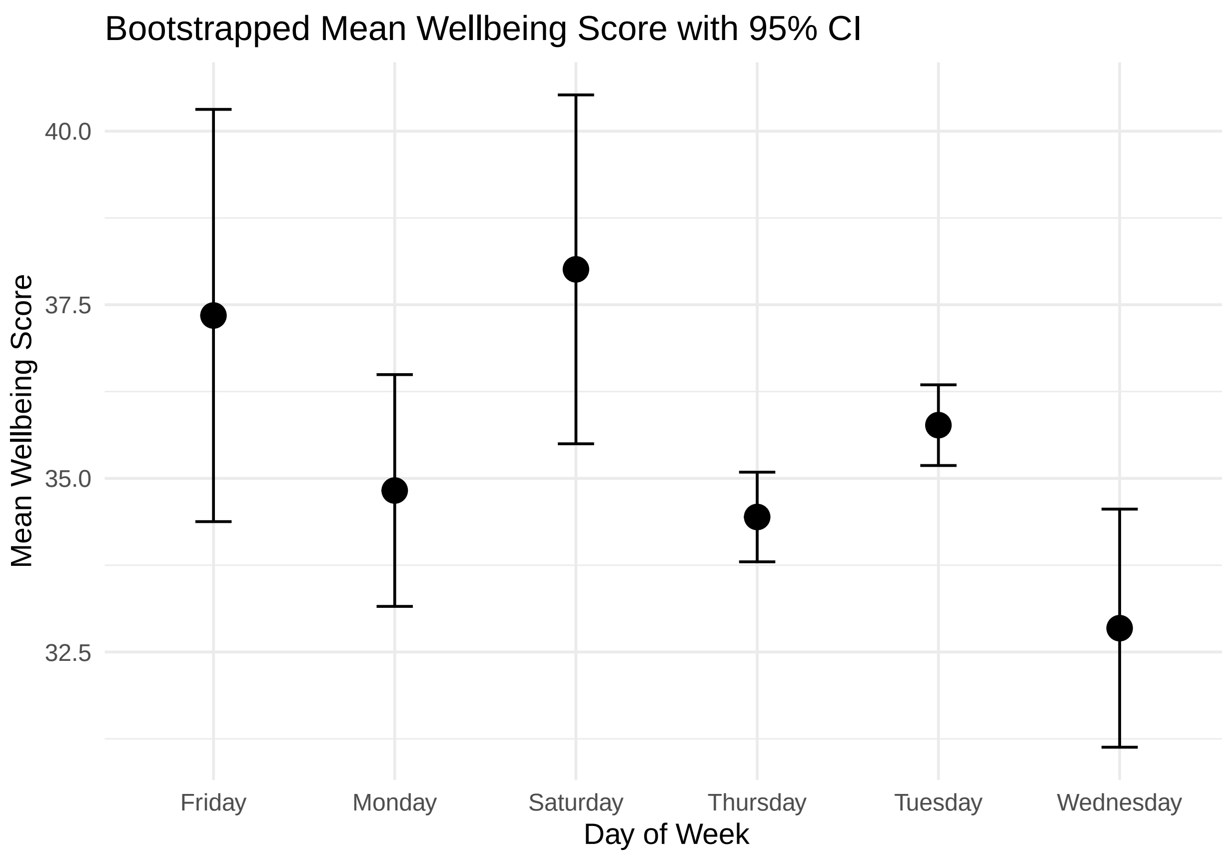
Implications

- Post-game recovery appears incomplete by Tuesday for some athletes, highlighting individual differences.
- Use Tuesday data to identify athletes who may need load modification or recovery support
- Use Thursday data to assess competition readiness and rebound quality.
- Targeted interventions can be designed based on these midweek insights to optimize performance and reduce injury risk.
- Individual variability midweek suggests value in personalized recovery strategies
- Flag athletes who fail to improve from Tuesday to Thursday for these interventions
- Add Friday survey to gauge Game day readiness.
- Sunday survey could assess post-game recovery dynamics.
- To determine practice readiness and recovery by using Specific readiness thresholds.

Visualization of Bootstrapped Mean Wellbeing Score with 95% Confidence Intervals The plot displays the average wellness score for each day of the week, along with 95% confidence intervals derived from bootstrapping. This graph conveys the overall average wellness scores of the team. This graph takes a macro look at our wellness scores. This is useful in identifying overall trends and setting team-level interventions.

Table 3: Average Wellbeing Score by Day of the Week

	Day_of_Week	‘Wellbeing Score‘	se
Friday	Friday	37.35	1.51
Monday	Monday	34.83	0.85
Saturday	Saturday	38.01	1.28
Thursday	Thursday	34.44	0.33
Tuesday	Tuesday	35.77	0.30
Wednesday	Wednesday	32.84	0.87



Confidence Interval Interpretation

- High response rate on Tuesday and Thursday yields narrow CIs, indicating precise estimates of average wellness on these days.
- Sparse data on other days results in wide CIs, reflecting uncertainty in average wellness estimates for those days.
- **Tuesday**
- Shows above-average wellness scores with a relatively narrow CI, suggesting consistent midweek recovery for many athletes.
- **Thursday**
- Lower average compared to Tuesday, this may indicate mid-week fatigue, training load effects, or a decrease in wellness as the week progresses.
- **Other Days**
- Wide CIs limit interpretability; more data would be needed to draw firm conclusions about average wellness on these days.

Implications

- **Tuesday** data should be our baseline health check day.
- **Thursday** data can inform adjustments to training loads or recovery strategies later in the week.
- An increase in **Friday** data collection could help assess pre-game readiness.
- An increase in **Sunday** data collection could help assess post-game recovery.

7. Regression Model: Predicting Wellbeing Score

A survey-weighted generalized linear regression model was fitted to examine the relationships between key wellness indicators (overall stress, physical fatigue, soreness, sleep quality, and motivation) and the overall wellness score.

Summary

- All five wellness indicators are statistically significant predictors of the overall wellness score ($p < 0.001$ for all coefficients).
- Motivation has the largest estimated effect. A one-unit increase in motivation is associated with an average 2.55-point increase in wellness
- Overall stress, physical fatigue, soreness, and sleep quality all have similarly sized effects, each contributing roughly 2.3–2.5 points to wellness per unit increase
- Sleep quality shows especially high precision, with the smallest standard error and the largest t-value, indicating a very stable relationship with wellness.
- The 95% confidence intervals are narrow and do not include zero for any predictor, reinforcing the reliability of the estimated effects
- Overall, the results indicate that wellness is strongly and consistently shaped by multiple, interrelated wellness factors, rather than by any single indicator in isolation

8. Interaction Model: Position Specific Effects

This model builds on our earlier analysis by recognizing that not every position responds to stress, fatigue, soreness, sleep, and motivation in the same way. Instead of assuming all players are affected equally, the model looks at how each position (Attack, Midfield, Defense, Goalie, Faceoff) responds differently to these wellness factors.

Summary

- All five wellness indicators—stress, fatigue, soreness, sleep quality, and motivation—are strongly associated with wellness across the team (all main effects $p < .001$)
- **Stress affects positions differently**
  - The impact of stress on wellness is significantly stronger for DM players
- **Physical fatigue shows mostly consistent effects across positions**
  - No strong evidence that fatigue impacts wellness differently by position
- **Soreness has position-specific importance**
  - DM and FO players experience a stronger negative impact of soreness on wellness
- **Sleep quality matters less for some positions**
  - The positive effect of sleep quality on wellness is reduced for OM, DM, and FO players compared to Attack, Defense, and Goalies
- **Motivation is especially critical for Defense players**

  - Motivation has a much stronger relationship with wellness for Defense than for any other position.

- **Overall**
  - Wellness factors do not affect all positions equally
  - Tailored interventions may need to address position and individual sensitivity to different wellness factors.

Practical Implications and Recommendations

9. Conclusion and Practical Applications

- Tuesday will anchor our weekly readiness framework.
- Thursday will inform adjustments to training loads and recovery strategies later in the week.
- Friday surveys could assess pre-game readiness.
- Sunday surveys could assess post-game recovery.
- Tuesday's wellness scores will be our baseline scores for the week.
- Changes in individual wellness scores from Tuesday to Thursday can flag athletes needing targeted interventions.
- Specific readiness thresholds can guide practice readiness and recovery strategies.
- Use findings to inform coaching strategies and optimize performance.
- Educate athletes on the importance of wellness monitoring and self-care practices.

9a. Long-Term Goals

- Integrate wellness data with training load metrics, Catapult and VALD, for comprehensive athlete monitoring. Specifically our Jump testing data, which is used to monitor neuromuscular fatigue and readiness.
- Using statistical analysis to observe the relationship between subjective wellness scores and objective measures. We want to test if they are portraying the same message about athlete readiness.
- Continuously refine wellness monitoring based on ongoing data collection and analysis.
- Continue to foster a culture of open communication around wellness to support athlete health and performance.
- Implement a machine learning–based predictive analytics framework to model how objective workload and neuromuscular metrics (CMJ, RSI, Catapult variables) relate to subjective wellness scores. The goal is to proactively flag athletes at elevated risk of reduced readiness and support informed load management decisions.

Appendicies

A. Bootstrap-Based Confidence Intervals for Wellness Indicators by Position

Table 4: Bootstrap-Based 95% Confidence Intervals for Wellness Indicators by Position

Position_Variable	Lower_95	Upper_95
A:Overall_Stress_Num	3.566	3.906
OM:Overall_Stress_Num	3.899	4.082
DM:Overall_Stress_Num	4.089	4.258
D:Overall_Stress_Num	3.841	4.064
GK:Overall_Stress_Num	3.901	4.198
FO:Overall_Stress_Num	3.421	3.787
A:Physical_Fatigue_Num	3.271	3.616
OM:Physical_Fatigue_Num	3.269	3.466
DM:Physical_Fatigue_Num	3.696	3.888
D:Physical_Fatigue_Num	3.593	3.856
GK:Physical_Fatigue_Num	3.343	3.673
FO:Physical_Fatigue_Num	3.018	3.359
A:Soreness_Num	3.373	3.702
OM:Soreness_Num	3.199	3.350
DM:Soreness_Num	3.611	3.770
D:Soreness_Num	3.450	3.716
GK:Soreness_Num	3.367	3.649
FO:Soreness_Num	3.446	3.782
A:Sleep_Quality_Num	3.439	3.768
OM:Sleep_Quality_Num	3.578	3.726
DM:Sleep_Quality_Num	3.646	3.886
D:Sleep_Quality_Num	3.786	4.025
GK:Sleep_Quality_Num	3.875	4.322
FO:Sleep_Quality_Num	2.901	3.297
A:‘Wellbeing Score‘	33.451	36.285
OM:‘Wellbeing Score‘	32.550	33.826
DM:‘Wellbeing Score‘	36.203	37.722
D:‘Wellbeing Score‘	35.113	37.139
GK:‘Wellbeing Score‘	35.185	37.536
FO:‘Wellbeing Score‘	31.409	33.502

B. Chi-Square Test - Wellness Indicators by Mlax_Position

Table 5: Physical Fatigue by Mlax Position

Mlax_Position	Physical.Fatigue	Freq
A	Very High	1
OM	Very High	1
DM	Very High	3
D	Very High	0
GK	Very High	0
FO	Very High	3
A	High	11
OM	High	45
DM	High	13
D	High	6
GK	High	3
FO	High	13
A	Average	51
OM	Average	148
DM	Average	135
D	Average	42
GK	Average	27
FO	Average	55
A	Low	26
OM	Low	76
DM	Low	97
D	Low	60
GK	Low	28
FO	Low	22
A	Very Low	17
OM	Very Low	43
DM	Very Low	98
D	Very Low	19
GK	Very Low	3
FO	Very Low	8

Table 6: Motivation by Mlax Position

Mlax_Position	Motivation_binary	Freq
A	Low	0
OM	Low	8
DM	Low	2
D	Low	0
GK	Low	1
FO	Low	2
A	Average	9

OM	Average	28
DM	Average	20
D	Average	0
GK	Average	0
FO	Average	9
A	High	6
OM	High	131
DM	High	79
D	High	34
GK	High	9
FO	High	15
A	Very High	91
OM	Very High	146
DM	Very High	245
D	Very High	93
GK	Very High	51
FO	Very High	75

Table 7: Wellbeing Category by Mlax Position

Mlax_Position	Wellbeing_cat	Freq
A	Low	50
OM	Low	159
DM	Low	99
D	Low	32
GK	Low	11
FO	Low	59
A	Medium	21
OM	Medium	81
DM	Medium	98
D	Medium	51
GK	Medium	23
FO	Medium	27
A	High	35
OM	High	73
DM	High	149
D	High	44
GK	High	27
FO	High	15

Table 8: Sleep Quality by Mlax Position

Mlax_Position	Sleep.Quality	Freq
A	Unable to Sleep	0
OM	Unable to Sleep	0
DM	Unable to Sleep	11

D	Unable to Sleep	0
GK	Unable to Sleep	0
FO	Unable to Sleep	1
A	Restless, Woke Up 2x +	3
OM	Restless, Woke Up 2x +	13
DM	Restless, Woke Up 2x +	40
D	Restless, Woke Up 2x +	2
GK	Restless, Woke Up 2x +	4
FO	Restless, Woke Up 2x +	30
A	Average	62
OM	Average	108
DM	Average	75
D	Average	29
GK	Average	10
FO	Average	40
A	Good, Feel Refreshed	15
OM	Good, Feel Refreshed	167
DM	Good, Feel Refreshed	113
D	Good, Feel Refreshed	75
GK	Good, Feel Refreshed	23
FO	Good, Feel Refreshed	18
A	Excellent, Feel Very Refreshed	26
OM	Excellent, Feel Very Refreshed	25
DM	Excellent, Feel Very Refreshed	107
D	Excellent, Feel Very Refreshed	21
GK	Excellent, Feel Very Refreshed	24
FO	Excellent, Feel Very Refreshed	12

Table 9: Overall Stress by Mlax Position

Mlax_Position	Overall_Stress_	Freq
A	High	5
OM	High	14
DM	High	6
D	High	1
GK	High	1
FO	High	6
A	Average	46
OM	Average	72
DM	Average	66
D	Average	26
GK	Average	6
FO	Average	40
A	Low	27
OM	Low	129
DM	Low	136

D	Low	78
GK	Low	43
FO	Low	40
A	Very Low	28
OM	Very Low	98
DM	Very Low	138
D	Very Low	22
GK	Very Low	11
FO	Very Low	15

Table 10: Soreness by Mlax Position

Mlax_Position	Soreness	Freq
A	Painful to Move	0
OM	Painful to Move	2
DM	Painful to Move	0
D	Painful to Move	0
GK	Painful to Move	0
FO	Painful to Move	0
A	Very Sore	11
OM	Very Sore	34
DM	Very Sore	5
D	Very Sore	11
GK	Very Sore	2
FO	Very Sore	3
A	Moderate	43
OM	Moderate	160
DM	Moderate	150
D	Moderate	45
GK	Moderate	26
FO	Moderate	50
A	Minimal	36
OM	Minimal	110
DM	Minimal	138
D	Minimal	57
GK	Minimal	33
FO	Minimal	31
A	No Soreness	16
OM	No Soreness	7
DM	No Soreness	53
D	No Soreness	14
GK	No Soreness	0
FO	No Soreness	17

C. Chi-Square Test - Wellness Indicators by MLAX Position - P-Values

Table 11: Rao–Scott Chi-square Test P-Value

Indicator	P_Value
Overall Stress	2.37485e-15

Table 12: Rao–Scott Chi-square Test P-Value

Indicator	P_Value
Physical Fatigue	1.559718e-11

Table 13: Rao–Scott Chi-square Test P-Value

Indicator	P_Value
Soreness	2.857674e-10

Table 14: Rao–Scott Chi-square Test P-Value

Indicator	P_Value
Sleep Quality	2.009205e-41

Table 15: Rao–Scott Chi-square Test P-Value

Indicator	P_Value
Motivation	4.17e-18

Table 16: Rao–Scott Chi-square Test P-Value

Indicator	P_Value
Overall Wellness	3.534137e-12

Chi-Square Test - Overall Stress

```
Chi_Stress_p <- as.numeric(chi_overall_stress$p.value)


chi_stress_pvalue <- data.frame(
  Indicator = c("Overall Stress"),
  P_Value = Chi_Stress_p,
  row.names = NULL
)
```

Summary

- **Overall Stress**
  - This suggests that there is a significant association between player position (*Mlax_Position*) and overall stress (*Overall_Stress_*). Certain positions may experience higher or lower levels of overall stress compared to others.

Chi-Square Test - Physical Fatigue

Summary

- **Physical Fatigue**
  - This suggests that there is a significant association between player position (*Mlax_Position*) and physical fatigue (*Physical Fatigue*). Certain positions may experience higher or lower levels of physical fatigue compared to others.

Chi-Square Test - Soreness

Summary

- **Soreness**
  - This suggests that there is a significant association between player position (*Mlax_Position*) and soreness (*Soreness_recode*). Certain positions may experience higher or lower levels of soreness compared to others.

Chi-Square Test - Sleep Quality

Summary

- **Sleep Quality**
  - This suggests that there is a significant association between player position (*Mlax_Position*) and sleep quality (*Sleep Quality*). Certain positions may experience better or poorer sleep quality compared to others.

Chi-Square Test - Motivation

Summary

- **Motivation**
  - This suggests that there is a significant association between player position (*Mlax_Position*) and motivation (*Motivation_binary*). Motivation levels appear to vary across positions.

Chi-Square Test - Overall Wellness

D. Chi-Square Test - Wellness Indicators by Wellbeing Score - P-Values

. ## Chi-Square Test - Overall Stress

Summary

The association between overall stress and wellness category is highly significant, indicating that stress levels vary meaningfully across low, medium, and high wellness groups.

Chi-Square Test - Physical Fatigue

Summary

Physical fatigue is also strongly related to wellness, suggesting that athletes reporting better wellness tend to experience systematically lower fatigue.

Chi-Square Test - Soreness

Summary

Soreness shows one of the strongest associations with wellbeing, indicating clear differences in physical soreness across wellbeing categories.

Chi-Square Test - Sleep Quality

Summary

Sleep quality is significantly associated with wellbeing, reinforcing the importance of sleep as a key contributor to athletes' overall wellness.

Chi-Square Test - Motivation

Summary

Motivation also differs substantially by wellbeing, with higher wellbeing linked to higher motivation

Table 17: Rao–Scott Chi-square Test P-Value

Indicator	P_Value
Overall Stress	3.822001e-101

Table 18: Rao–Scott Chi-square Test P-Value

Indicator	P_Value
Physical Fatigue	1.76832e-121

Table 19: Rao–Scott Chi-square Test P-Value

Indicator	P_Value

Soreness	4.92772e-105

Table 20: Rao–Scott Chi-square Test P-Value

Indicator	P_Value
Sleep Quality	3.880394e-96

Table 21: Rao–Scott Chi-square Test P-Value

Indicator	P_Value
Motivation	2.499099e-46

E. Methodology

1. Bootstrapping A method that repeatedly re-samples the data to evaluate how stable the results are. This allows confidence ranges to be attached to averages and model estimates. With these steps in place, the analysis compares wellness indicators across positions, tests whether patterns differ meaningfully by position or wellness level, and examines how stress, fatigue, soreness, sleep, and motivation jointly relate to overall wellness.

Bootstrapping is used to measure uncertainty in the results by repeatedly re-analyzing the same survey data 1,000 times. Each repetition mimics what could happen if a slightly different group of players responded, which helps capture natural variability caused by who chose to complete the survey. This approach is especially useful when the entire team is invited to participate but not everyone responds. Using this method, average levels of stress, fatigue, soreness, sleep quality, and overall wellness are calculated for each position. Statistical uncertainty is then quantified using 95% confidence intervals, which provide a reasonable range where the true average values are likely to fall.

2. Use of the Survey Package in R Using a statistical adjustment process, responses from positions that participated more than expected are given slightly less influence, while responses from positions with lower participation are given more influence. This produces a new set of position-adjusted weights for each survey response.

These adjusted weights are then applied to create the final survey structure that will be used for all remaining analyses. From this point forward, every average, comparison, test, and model reflects the true positional makeup of the team rather than raw response counts.

3. Chi Square testing To make fair comparisons, several wellness measures are first simplified into broader categories (for example, combining “Very High” and “High” stress into a single “High” category). This ensures there are enough responses in each group to support reliable comparisons across positions.

The analysis then uses chi-square tests, a statistical method designed to evaluate whether two categorical variables are related. In this context, a chi-square test asks a simple question: Are differences in wellness (such as stress, fatigue, soreness, sleep quality, motivation, or overall wellness level) distributed evenly across positions, or do some positions experience these outcomes more often than others? If the differences observed are larger than what would reasonably occur by chance, the test indicates a meaningful association.

Importantly, these chi-square tests are adjusted for the survey design, meaning they account for unequal response rates and differences in position size. This ensures the results reflect true positional patterns rather than artifacts of who happened to respond more frequently.

Together, these analyses identify which aspects of wellness are linked to position-specific demands, providing evidence-based insight into how stress, recovery, sleep, motivation, and overall wellness vary across the team.

This information can be used to guide targeted training loads, recovery strategies, and support interventions tailored to different positional roles.

4. Regression modeling The model accounts for the complex survey design using bootstrap weights to ensure accurate standard errors and confidence intervals.

5. Interaction modeling This model extends the baseline survey-weighted regression by including interaction terms between wellness indicators and playing position, allowing the effects of stress, fatigue, soreness, sleep quality, and motivation on wellness to vary by position. Rather than assuming homogeneous effects across athletes, the model estimates position-specific slopes, capturing differential sensitivity to wellness factors that would be obscured in a pooled model. Estimation uses bootstrap-based survey weights to ensure valid inference under the complex survey design.